

Design Calculations						
Calculation						
Component	Thickness (m)	Porosity	Density (kg/m³)	Area (m²)	Mass (kg)	Mass (g)
Negative electrode	0.0001	0.32	3260	0.1027	=B3*(1-C3)*D3*E3	=F3*1000
Negative CC	1e-05	0	2700	0.1027	=B4*(1-C4)*D4*E4	=F4*1000
Separator	1.5e-05	0.45	900	0.1027	=B5*(1-C5)*D5*E5	=F5*1000
Positive CC	5e-06	0	8960	0.1027	=B6*(1-C6)*D6*E6	=F6*1000
Positive electrode	9e-05	0.28	1660	0.1027	=B7*(1-C7)*D7*E7	=F7*1000
Total					=SUM(F3:F7)	=SUM(G3:G7)
Energy Density Calculation						
Parameter	Value	Unit	Source			
Charge energy	19.367	Wh	r3_archF_1c_energy.json:energy_wh			
Mass	0.04142	kg	Sum of layer masses			
Energy density	=B12/B13	Wh/kg	Energy / Mass			
Volumetric ED	=B12/0.0000226	Wh/L	Energy / Volume			
Midpoint voltage	3.993	V	r3_archF_1c_energy.json:midpoint_voltage_v			
Resistance	0.000115	Ω	r3_archF_1c_energy.json:dcr_ohm			
Power density	=B16^2/(4*B17)/B13	W/kg	V_OC²/(4·R) / mass			
Thermal Capability						
Condition	Capacity (Ah)	Retention (%)	T_max (K)	Source		
Discharge	5.326	100	308.8	r3_archF_1c.json		
Charge	5.071	=B22/B22*100	386.6	r3_archF_5c.json		
Retention	=B23/B22	target>=90%				
Stability Assessment						
	T_max (K)	Limit (K)	Verdict	Anode min (V)	Plating	
Charge 45°C	361.8	358.15	MARGINAL	0.03	No	